

DBMS Assignment 3 – Local Case Study

Local Case Study: Kirana / General Store

1. Current Method Summary:

The selected business is **Shanti Kirana Store**, a local Kirana/General Store that sells daily-use products such as groceries, snacks, beverages, personal-care products, and household items. The store maintains records related to daily sales, product stock, customer orders, supplier details, and customer credit or dues.

Currently, the store mainly uses a notebook and paper bills to maintain its records. Daily sales are written in a sales register, while information about customer dues is maintained separately in a notebook. Stock is checked manually by looking at the available products on the shelves and in the storage area. Supplier details and purchase information are also maintained through handwritten records and bills. Some regular customer orders may be received through phone calls or WhatsApp messages.

The owner and staff members access and update these records whenever required. Since most of the information is maintained manually, finding older records or checking the exact stock and outstanding customer dues can take additional time.

2. Flaws & Limitations:

The current manual method has several limitations. Maintaining information in different notebooks and paper bills can result in duplicate entries and data redundancy. There can also be inconsistencies if the same information is recorded differently in different places.

Paper records can be lost, damaged, or misplaced, and maintaining a backup is difficult. Searching for an old bill or a previous transaction can also be time-consuming. Manually checking stock may result in delays in identifying products that need to be reordered. Similarly, tracking customer credit and pending payments becomes difficult when the information is spread across handwritten records.

There are also security concerns because physical notebooks and bills do not provide proper user-based access control. If more than one staff member updates records, there can be a possibility of incorrect or conflicting entries.

3. Proposed DBMS Solution & Component Mapping:

A relational DBMS can organize the store's information in structured tables such as **Products, Sales, Customers, Suppliers, Orders, and Customer Dues**. These tables can be related to each other so that information can be stored and retrieved efficiently.

The **Query Processor** can help the owner search for specific information, such as previous sales, customer dues, available stock, or supplier records.

The **Storage Manager** can manage the storage and retrieval of the store's sales, inventory, customer, supplier, and order information.

The **Buffer Manager** can temporarily manage data while records are being accessed or processed during operations such as billing and stock checking.

The **Transaction Manager** can ensure that operations such as completing a sale and updating the corresponding stock are handled consistently.

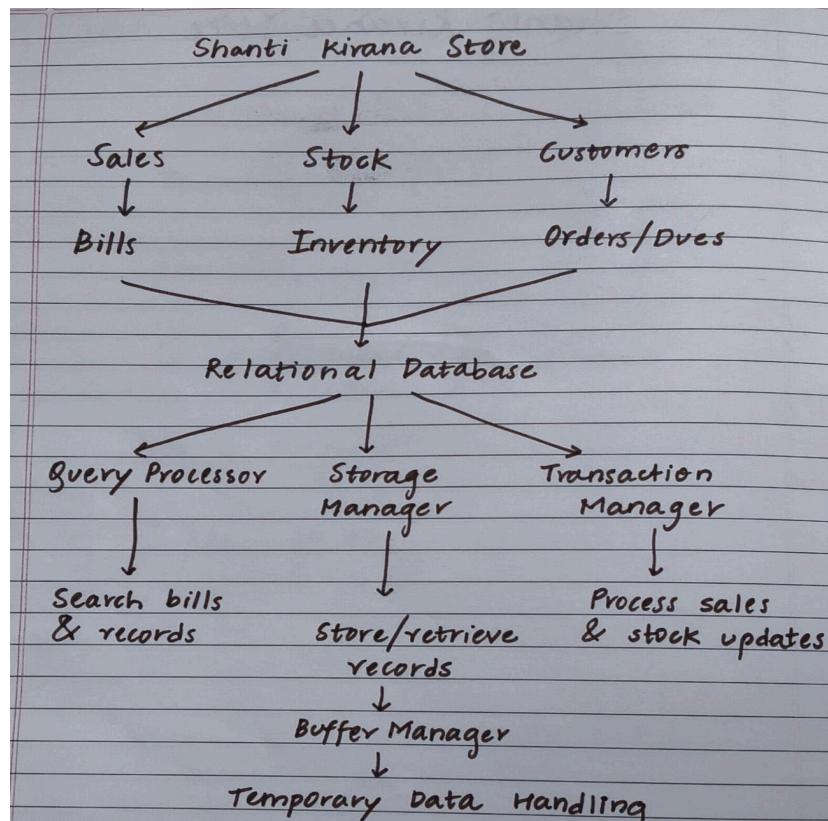
The **DDL/DML Interpreter** can be used to create database structures such as product and sales tables and to perform operations such as inserting, updating, and deleting records.

The **Authorization/Security Subsystem** can control access to the database. For example, the owner can have full access while staff members can be given access only to the information required for their work.

Thus, a relational DBMS can provide the store with organized data storage, faster searching, better consistency, controlled access, and more reliable management of sales, stock, customers, suppliers, and dues.

4. Hand-Drawn DBMS Component Diagram:

The hand-drawn diagram maps the physical operations of **Shanti Kirana Store** to DBMS components.



Authorization / security

Owner / staff

DDL / DML Interpreter

Create / Insert / update / Delete records

Annexure – Interview Notes

Business Details:

Business Type: Kirana / General Store

Business Name: Shanti Kirana Store

Person Interviewed: RaghuDada (Shopkeeper)

Interview Questions and Answers:

Q1. What records do you maintain in your shop?

Answer: We maintain records of daily sales, stock, purchases from suppliers, customer orders, and payments or dues from regular customers.

Q2. How do you maintain your daily sales records?

Answer: Daily sales are mainly recorded in a notebook or sales register. Bills are also kept for some transactions.

Q3. How do you keep track of your stock/inventory?

Answer: Stock is mostly checked manually by looking at the products available in the shop and storage area. When an item is running low, it is added to the purchase list.

Q4. How do you maintain customer orders?

Answer: Regular customers sometimes give orders directly in the shop or through phone calls and WhatsApp messages. The details are written down when necessary.

Q5. How do you maintain supplier details?

Answer: Supplier names, contact details, purchase information, and bills are maintained using notebooks and physical bills.

Q6. Do you provide products on credit? If yes, how do you track customer dues?

Answer: Yes, some regular customers purchase items on credit. Their names and pending amounts are written in a separate notebook and updated when they make payments.

Q7. Where do you store these records?

Answer: Most records are stored in notebooks, registers, and physical bills. Some order information is also available through WhatsApp messages.

Q8. Who accesses and updates the records?

Answer: The owner mainly checks and updates the records. Staff members also handle sales and may update some entries when required.

Q9. What problems do you face while maintaining these records?

Answer: It can be difficult to find older records quickly. Sometimes entries can be missed or written incorrectly. Keeping track of stock and customer dues also requires regular manual checking.

Q10. Is it difficult to find old bills or previous records?

Answer: Yes. If the record is old, we sometimes have to search through notebooks and physical bills to find the required information.

Q11. How do you correct mistakes in the records?

Answer: Mistakes in handwritten records are usually corrected manually by crossing out the incorrect entry and writing the correct information.

Q12. Do you have a backup of your records?

Answer: There is no complete digital backup of all the handwritten records. Physical notebooks and bills are kept safely in the shop.